

Tab 1

EXPERIMENT 11

Aim - Implementation of Hyperledger Fabric Network and Asset Management using Chaincode.

Theory -

Hyperledger Fabric Network Setup

- Consortium & Membership Service Provider (MSP): Fabric is permissioned, meaning only authorized organizations can join. MSPs define identities and roles.
- Peers: Nodes that maintain the ledger and endorse transactions. Each organization typically runs peers.
- Orderer: The ordering service packages endorsed transactions into blocks and delivers them to peers.
- Ledger: Composed of a blockchain (immutable transaction log) and a state database (current values).
- Channels: Private sub-networks where a subset of organizations transact confidentially.

Chaincode (Smart Contracts)

- Chaincode defines the business logic for asset management.
- Written in Go, Java, or Node.js.
- Deployed to peers, then instantiated on a channel.
- Functions typically include createAsset, queryAsset, transferAsset, and deleteAsset.

Transaction Flow

1. Proposal Creation: A client application submits a transaction proposal (e.g., create asset).
2. Endorsement: Peers simulate the transaction using chaincode and return signed results.
3. Ordering: The orderer sequences endorsed transactions into blocks.
4. Validation & Commit: Peers validate endorsements, check for conflicts, and commit the block to the ledger.

Asset Operations

- Asset Creation:
 - Chaincode function createAsset stores asset attributes (ID, owner, value) in the ledger's state database.
- Asset Querying:
 - queryAsset retrieves asset details by key. Rich queries are possible if CouchDB is used as the state database.
- Asset Transfer:
 - transferAsset updates the owner field of an asset. The transaction is endorsed, ordered, and committed, ensuring immutability and consensus.

Key Principles

- Immutability: Once committed, transactions cannot be altered.

- Consensus: Endorsement policies ensure multiple organizations agree before a transaction is valid.
- Privacy: Channels and private data collections allow selective visibility.
- Trust: Identities are cryptographically verified via MSPs.

In essence, Hyperledger Fabric provides a modular, permissioned blockchain framework where chaincode governs asset lifecycles. The theory emphasizes identity-driven trust, modular architecture, and consensus-based validation — ensuring secure creation, querying, and transfer of assets across organizations.

Prerequisite: Check if WSL is Installed

Open **PowerShell (Run as Administrator)** and run:

wsl --list --verbose

If it shows **no distributions**, you must install Ubuntu.

Install Ubuntu for WSL

Run this command in **PowerShell (Admin)**:

wsl --install -d Ubuntu

This installs:

- Ubuntu
- WSL2 kernel

The installation may take a few minutes.

Restart Your Computer

After installation finishes, **restart your PC**.

Open Ubuntu

Open:

Start Menu → Ubuntu

It will ask you to create:

Username

Password

Now Ubuntu is ready.

Implementation

Step 1 — Update the system

code :

sudo apt update

sudo apt upgrade -y

Explain: `apt update` refreshes the package index — it tells your system what versions of software are available. `apt upgrade` installs the newer versions. Fabric's dependencies (Docker, curl, git) can fail or behave unexpectedly on outdated system libraries. Always do this first on a fresh machine.

```
pranav@DESKTOP-OSHSLJI:~$ sudo apt update
sudo apt upgrade -y
[sudo] password for pranav:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1540 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [246 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.6 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [976 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [219 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.8 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2683 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [623 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [208 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6732 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:22 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1837 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [336 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
```

Step 2 — Verify prerequisites

code :

```
sudo apt install jq curl git nodejs npm -y
```

```
node -v
```

```
npm -v
```

```
git --version
```

```
docker --version
```

```
docker ps
```

```
pranav@DESKTOP-OSHSLJI:~$ node -v
v18.19.1
pranav@DESKTOP-OSHSLJI:~$ npm -v
11.8.0
pranav@DESKTOP-OSHSLJI:~$ git --version
git version 2.43.0
pranav@DESKTOP-OSHSLJI:~$ docker --version
docker ps
Docker version 29.2.0, build 0b9d198
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
--------------	-------	---------	---------	--------	-------	-------

Explain: Fabric requires Docker (to run peer/orderer containers), Git (to clone the repo), and Node.js + npm (to build JavaScript chaincode). Running `docker ps` with no output confirms the Docker daemon is running and your user can access it. If `docker ps` requires sudo, you need to add your user to the docker group.

Step 3 — Clone fabric-samples

code :

```
git clone https://github.com/hyperledger/fabric-samples
```

```
cd fabric-samples
```

```

pranav@DESKTOP-OSHSLJI:~$ git clone https://github.com/hyperledger/fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15527, done.
remote: Counting objects: 100% (98/98), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 15527 (delta 57), reused 38 (delta 38), pack-reused 15429 (from 3)
Receiving objects: 100% (15527/15527), 24.16 MiB | 6.18 MiB/s, done.
Resolving deltas: 100% (8722/8722), done.
pranav@DESKTOP-OSHSLJI:~$ cd fabric-samples

```

Explain: `fabric-samples` is the official repository containing the test-network scripts, ready-made chaincode examples (including `asset-transfer-basic`), and configuration files. Everything else in this guide runs from inside this directory.

Step 4 — Download the install script and run it

code :

`curl -sLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh`

`chmod +x install-fabric.sh`

`./install-fabric.sh docker samples binary`

`./install-fabric.sh docker`

```

pranav@DESKTOP-OSHSLJI:~/fabric-samples$ curl -sLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
pranav@DESKTOP-OSHSLJI:~/fabric-samples$ chmod +x install-fabric.sh
pranav@DESKTOP-OSHSLJI:~/fabric-samples$ ./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

====> Downloading version 2.5.15 platform specific fabric binaries
====> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz
====> Will unpack to: /home/pranav/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0  0:00:00  0:00:00  0:00:00     0
100 120M  100 120M    0     0 6272k      0  0:00:19  0:00:19  0:00:00 6483k
==> Done.
====> Downloading version 1.5.17 platform specific fabric-client binary

```

```

====> List out hyperledger images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15      654bd4554bf9      250MB      80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17          453a0a727e82      381MB      123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15       86ddd7eda92       1GB        255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15     9972c209be47      178MB      49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15        3e25adc31a75      230MB      66.2MB
hyperledger/fabric-baseos:2.5                 654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:latest              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-ca:1.5                     453a0a727e82      381MB      123MB
hyperledger/fabric-ca:1.5.17                  453a0a727e82      381MB      123MB
hyperledger/fabric-ca:latest                   453a0a727e82      381MB      123MB
hyperledger/fabric-ccenv:2.5                  86ddd7eda92       1GB        255MB
hyperledger/fabric-ccenv:2.5.15               86ddd7eda92       1GB        255MB
hyperledger/fabric-ccenv:latest               86ddd7eda92       1GB        255MB
hyperledger/fabric-orderer:2.5                9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:2.5.15             9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:latest             9972c209be47      178MB      49.5MB
hyperledger/fabric-peer:2.5                   3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:2.5.15                3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:latest                3e25adc31a75      230MB      66.2MB

```

```

pranav@DESKTOP-OSHSLJI:~/fabric-samples$ ./install-fabric.sh docker

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
====> Pulling fabric Images
====> ghcr.io/hyperledger/fabric-peer:2.5.15
2.5.15: Pulling from hyperledger/fabric-peer
Digest: sha256:3e25adc31a757ee19dd8a24afdc9ac5aa46a86b3cb56fc93068b0e594706d6aa
Status: Image is up to date for ghcr.io/hyperledger/fabric-peer:2.5.15
ghcr.io/hyperledger/fabric-peer:2.5.15
====> ghcr.io/hyperledger/fabric-orderer:2.5.15
2.5.15: Pulling from hyperledger/fabric-orderer
Digest: sha256:9972c209be47f45e377a24c88f2e3d21fae4dc224751c68bdd33f2e7a603ffa8
Status: Image is up to date for ghcr.io/hyperledger/fabric-orderer:2.5.15
ghcr.io/hyperledger/fabric-orderer:2.5.15
====> ghcr.io/hyperledger/fabric-ccenv:2.5.15
2.5.15: Pulling from hyperledger/fabric-ccenv
Digest: sha256:86dded7eda9268e2cbf3691cb952edf545dc5faea1a79bfc9b47ec47d3ecfa9e
Status: Image is up to date for ghcr.io/hyperledger/fabric-ccenv:2.5.15
ghcr.io/hyperledger/fabric-ccenv:2.5.15

```

Explain: This one script does three things simultaneously. **docker** pulls all Fabric Docker images — **fabric-peer**, **fabric-orderer**, **fabric-ccenv**, **fabric-baseos**, **fabric-ca**, and critically **fabric-nodeenv** (needed to build Node.js chaincode). **binary** downloads the CLI tools — **peer**, **configtxgen**, **configtxlator** — into **~/fabric-samples/bin**. **samples** ensures the chaincode sample directories are present. Without **fabric-nodeenv**, chaincode deployment fails later.

Step 5 — Verify Docker images are present

code :

docker images | grep fabric

```

pranav@DESKTOP-OSHSLJI:~/fabric-samples$ docker images | grep fabric
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15      654bd4554bf9      250MB      80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17         453a0a727e82      381MB      123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15       86dded7eda92      1GB        255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15     9972c209be47      178MB      49.5MB      U
ghcr.io/hyperledger/fabric-peer:2.5.15        3e25adc31a75      230MB      66.2MB      U
hyperledger/fabric-baseos:2.5                 654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:2.5.15              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:latest              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-ca:1.5                     453a0a727e82      381MB      123MB
hyperledger/fabric-ca:1.5.17                  453a0a727e82      381MB      123MB
hyperledger/fabric-ca:latest                  453a0a727e82      381MB      123MB
hyperledger/fabric-ccenv:2.5                  86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:2.5.15               86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:latest               86dded7eda92      1GB        255MB
hyperledger/fabric-nodeenv:2.5                17e2d447ca0d      570MB      146MB
hyperledger/fabric-orderer:2.5                9972c209be47      178MB      49.5MB      U
hyperledger/fabric-orderer:2.5.15             9972c209be47      178MB      49.5MB      U
hyperledger/fabric-orderer:latest             9972c209be47      178MB      49.5MB      U
hyperledger/fabric-peer:2.5                   3e25adc31a75      230MB      66.2MB      U
hyperledger/fabric-peer:2.5.15                3e25adc31a75      230MB      66.2MB      U
hyperledger/fabric-peer:latest                3e25adc31a75      230MB      66.2MB      U

```

Explain: Confirm **hyperledger/fabric-nodeenv:2.5** is in the list. This image is the Node.js build environment that Fabric uses to compile and run JavaScript chaincode inside a container. If it is missing, **deployCC** will fail with a "No such image" error.

Step 6 — Set PATH and config path

code :

export PATH=\$PATH:\$HOME/fabric-samples/bin

export FABRIC_CFG_PATH=\$HOME/fabric-samples/config

```

pranav@DESKTOP-OSHSLSJI:~/fabric-samples$ export PATH=$PATH:$HOME/fabric-samples/bin
pranav@DESKTOP-OSHSLSJI:~/fabric-samples$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
pranav@DESKTOP-OSHSLSJI:~/fabric-samples$ peer version
peer:
Version: v2.5.15
Commit SHA: 83c7930
Go version: go1.26.0
OS/Arch: linux/amd64
Chaincode:
Base Docker Label: org.hyperledger.fabric
Docker Namespace: hyperledger

```

Explain: The `peer` binary lives in `~/fabric-samples/bin`, which is not on your shell's PATH by default. Without this export, the shell cannot find the command and returns "peer not found". `FABRIC_CFG_PATH` tells the peer CLI where to find `core.yaml`, the main configuration file. Without it, the peer errors with "cannot init crypto, path does not exist".

code :

peer version

You should see **Version: v2.5.15**.

Step 7 — Start the network and create the channel

code :

cd ~/fabric-samples/test-network

./network.sh up createChannel

```

pranav@DESKTOP-OSHSLSJI:~/fabric-samples/test-network$ cd ~/fabric-samples/test-network
./network.sh down
Using docker and docker-compose
Stopping network
[+] down 7/7
✓Container peer0.org1.example.com      Removed
✓Container orderer.example.com         Removed
✓Container peer0.org2.example.com      Removed
✓Network fabric_test                  Removed
✓Volume compose_orderer.example.com    Removed
✓Volume compose_peer0.org1.example.com Removed
✓Volume compose_peer0.org2.example.com Removed
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59f864cfa1398b52d341cb:latest
Deleted: sha256:27056269ed9759828700d7bfc21149ac76c17d725ae6f9a12c07cc42a04f539f
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59163835f604aa885d60dc:latest
Deleted: sha256:f7c74fe5fd5a6a73117e649e923a5de0a595771f793ee8e1b0047fc35d2d315c

```

```

pranav@DESKTOP-OSHSLJ1:~/fabric-samples/test-network$ ./network.sh up createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using data
Bringing up network
LOCAL_VERSION=v2.5.15
DOCKER_IMAGE_VERSION=v2.5.15
/home/pranav/fabric-samples/test-network/./bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] up 7/7
✓Network fabric_test Created
✓Volume compose_orderer.example.com Created
✓Volume compose_peer0.org1.example.com Created
✓Volume compose_peer0.org2.example.com Created
✓Container peer0.org1.example.com Created
✓Container orderer.example.com Created
✓Container peer0.org2.example.com Created
CONTAINER ID IMAGE COMMAND CREATED STATUS

```

Explain: This single command does everything needed to have a running network with a channel. Internally it uses **cryptogen** to generate TLS certificates and identity material (MSP) for Org1, Org2, and the Orderer. It starts three Docker containers — **orderer.example.com** on port 7050, **peer0.org1.example.com** on port 7051, and **peer0.org2.example.com** on port 9051. It then runs **configtxgen** to create **mychannel.block** (the channel genesis block), submits it to the orderer, joins both peers to the channel, and sets anchor peers so the two organizations can discover each other via gossip. The default channel name is **mychannel**.

Step 8 — Deploy the chaincode

code :

```

./network.sh deployCC \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccl javascript

```

```

pranav@DESKTOP-OSHSLJ1:~/fabric-samples/test-network$ ./network.sh deployCC \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript \
-ccl javascript
Using docker and docker-compose
Deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --lang
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1

```



```

2026-03-17 16:28:58.033 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [b424caca912525c6970bb1da73f421e25a00030499eab767783fa7e
atus (VALID) at localhost:7051
2026-03-17 16:28:58.037 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [b424caca912525c6970bb1da73f421e25a00030499eab767783fa7e
atus (VALID) at localhost:9051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required

```

Explain: `-ccn basic` is the name your chaincode will be called by in all future commands. `-ccp` is the path to the chaincode source code. `-ccl javascript` tells Fabric to use the Node.js runtime. Internally this runs the full Fabric chaincode lifecycle — package the source into a `.tar.gz`, install it on both peers, have Org1 approve the definition, have Org2 approve the definition (both approvals are required since majority endorsement is the default policy), and finally commit the definition to the channel. Once committed, two new Docker containers appear — one chaincode container per peer (`dev-peer0.org1...` and `dev-peer0.org2...`).

Step 9 — Set peer identity environment variables

code :

```

export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export
CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export
CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051

```

```

pranav@DESKTOP-OSHSLJ1:~/fabric-samples/test-network$ export PATH=$PATH:$HOME/fabric-samples/bin
export FABRIC_CFG_PATH=$HOME/fabric-samples/config

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051

```

Explain: The `peer` CLI has no built-in session — it does not know who you are. Every command needs to know which organization's identity to use, which peer to connect to, and how to verify TLS. These five variables provide exactly that. `CORE_PEER_LOCALMSPID` declares "I am Org1". `CORE_PEER_MSPCONFIGPATH` points to the Admin user's private signing key and certificate. `CORE_PEER_TLS_ROOTCERT_FILE` provides the CA certificate to verify the

peer's TLS identity. **CORE_PEER_ADDRESS** is which peer to send requests to. These must be re-exported in every new terminal session.

Step 10 — Initialize the ledger with sample data

code :

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/  
orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/  
peer0.org1.example.com/tls/ca.crt \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/  
peer0.org2.example.com/tls/ca.crt \  
--waitForEvent \  
-c '{"function": "InitLedger", "Args": []}'
```

```
pranav@DESKTOP-0SHSLJ1:~/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscac  
.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.co  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.co  
--waitForEvent \  
-c '{"function": "InitLedger", "Args": []}'  
2026-03-17 16:37:35.757 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [f982da882feb2411792ddb2c62b47591512c0269f789efe8ab9ce462  
atus (VALID) at localhost:9051  
2026-03-17 16:37:35.757 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [f982da882feb2411792ddb2c62b47591512c0269f789efe8ab9ce462  
atus (VALID) at localhost:7051  
2026-03-17 16:37:35.758 UTC 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
```

Explain: **invoke** is used for any operation that writes to the ledger. **-o localhost:7050** is the orderer address. **--tls** and **--cafile** enable TLS and provide the CA cert to verify the orderer. **-C mychannel** and **-n basic** identify which channel and chaincode. The two **--peerAddresses** and **--tlsRootCertFiles** pairs are the critical part — the default endorsement policy requires a majority of organizations to endorse. With two orgs, both must sign. Sending to only one peer results in **ENDORSEMENT_POLICY_FAILURE**. **--waitForEvent** blocks until both peers confirm the

transaction is committed (status VALID) rather than just submitted. **InitLedger** seeds the ledger with six sample assets.

Step 11 — Query all assets

code :

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'
```

```
pranav@DESKTOP-OSHSLSJI:~/fabric-samples/test-network$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'  
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
```

Explain: **query** is for read-only operations. No transaction is created, no block is written, no endorsement from multiple peers is needed. It is fast and cheap. This returns all six assets seeded by **InitLedger**:

```
[{"ID":"asset1","Color":"blue","Size":5,"Owner":"Tomoko","AppraisedValue":300},  
 {"ID":"asset2","Color":"red","Size":5,"Owner":"Brad","AppraisedValue":400},  
 {"ID":"asset3","Color":"green","Size":10,"Owner":"Jin Soo","AppraisedValue":500},  
 {"ID":"asset4","Color":"yellow","Size":10,"Owner":"Max","AppraisedValue":600},  
 {"ID":"asset5","Color":"black","Size":15,"Owner":"Adriana","AppraisedValue":700},  
 {"ID":"asset6","Color":"white","Size":15,"Owner":"Michel","AppraisedValue":800} ]
```

Step 12 — Read a specific asset

code :

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset5"]}'
```

```
pranav@DESKTOP-OSHSLSJI:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset5"]}'  
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}
```

Result:

```
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15}
```

Explain: **ReadAsset** takes an asset ID and returns that single asset's data from the world state. The world state is a key-value database that stores the current (latest) value for each asset. The full transaction history is stored in the blockchain, but queries read from this faster world state.

Step 13 — Create a new asset

code :

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/  
orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/  
peer0.org1.example.com/tls/ca.crt \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/  
peer0.org2.example.com/tls/ca.crt \  
--waitForEvent \  
-c '{"function":"CreateAsset","Args":["asset10","blue","10","Arnav","1000"]}'
```

```
pranav@DESKTOP-OSHSLSJI:~/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tls  
.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example  
--waitForEvent \  
-c '{"function":"CreateAsset","Args":["asset10","blue","10","Pranav","1000"]}'  
2026-03-17 16:38:30.370 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [d88abbf8b7aaff8416fb4a8914a4d01c8064d053ebc5145787034]  
atus (VALID) at localhost:7051  
2026-03-17 16:38:30.371 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [d88abbf8b7aaff8416fb4a8914a4d01c8064d053ebc5145787034]  
atus (VALID) at localhost:9051  
2026-03-17 16:38:30.371 UTC 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200  
{"Color":"blue","Size":10,"Owner":"Pranav","AppraisedValue":1000}
```

Explain: **CreateAsset** takes five arguments in order — [ID, Color, Size, Owner, AppraisedValue]. This is a write operation so both peers must endorse. The orderer packages the endorsed transaction into a block and delivers it to both peers. Both confirm VALID and the asset is permanently recorded on the ledger.

code :

```
peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
```

```
pranav@DESKTOP-OSHSLSJI:~/fabric-samples/test-network$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset10"]}'  
{"AppraisedValue":1000,"Color":"blue","ID":"asset10","Owner":"Pranav","Size":10}
```

```
# {"AppraisedValue":1000,"Color":"blue","ID":"asset10","Owner":"Arnav","Size":10}
```

The transaction flow showing what happens inside Fabric every time you run **invoke** or **query** Shows what happens inside Fabric every time you run **peer chaincode invoke** — the proposal, endorsement, ordering, and commit flow — and why both peers are always required for any write operation.

Step 14 — Transfer asset ownership

code :

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/  
orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/  
peer0.org1.example.com/tls/ca.crt \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/  
peer0.org2.example.com/tls/ca.crt \  
--waitForEvent \  
-c '{"function": "TransferAsset", "Args": ["asset10", "Rahul"]}'
```

```
pranav@DESKTOP-OSHSLJI:~/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.c  
.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \  
--waitForEvent \  
-c '{"function": "TransferAsset", "Args": ["asset10", "Rahul"]}'  
2026-03-17 16:41:28.692 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [e834ab1020eac67d6bbf5759291d3c3bd92083d1a340d1f5bd35104c2e387019] committed  
atus (VALID) at localhost:9051  
2026-03-17 16:41:28.692 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [e834ab1020eac67d6bbf5759291d3c3bd92083d1a340d1f5bd35104c2e387019] committed  
atus (VALID) at localhost:7051  
2026-03-17 16:41:28.692 UTC 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Pranav"
```

Explain: **TransferAsset** takes [**assetID**, **newOwner**]. It reads the current asset from the world state, updates the Owner field, and writes it back. The function returns the previous owner ("**Arnav**") as confirmation. Both peers endorse, the orderer commits the block, and the ledger is updated permanently.

Verify the ownership change:

code :

```
peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
```

```
# {"AppraisedValue":1000,"Color":"blue","ID":"asset10","Owner":"Rahul","Size":10}
```

```
2026-03-17 16:41:28.692 UTC 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Pranav"
pranav@DESKTOP-OSHSLJI:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
{"AppraisedValue":1000,"Color":"blue","ID":"asset10","Owner":"Rahul","Size":10}
```

Owner is now **Rahul**

Step 15 — To stop/bring down the Hyperledger Fabric network, run:

[./network.sh](#) down

```
pranav@DESKTOP-OSHSLJI:~/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 7/7
✓Container orderer.example.com          Removed
✓Container peer0.org2.example.com        Removed
✓Container peer0.org1.example.com        Removed
✓Network fabric_test                    Removed
✓Volume compose_peer0.org1.example.com  Removed
✓Volume compose_peer0.org2.example.com  Removed
✓Volume compose_orderer.example.com     Removed
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035f864cfa1398b52d341cb:latest
Deleted: sha256:aa29d3b9fb04e6316131fbf334a23f28d8a1646fa9d40090916c02d54bc5b76d
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035163835f604aa885d60dc:latest
Deleted: sha256:08d49c295856f950367880744bcacfaf56a7134bf208cacfab8a29ef9221703c
```

Conclusion

The Hyperledger Fabric network was successfully implemented, demonstrating asset creation, querying, and transfer using chaincode. The process highlights key concepts such as consensus, immutability, and secure transaction flow. Overall, it provides a clear understanding of permissioned blockchain systems and their real-world applications in asset management.

Tab 2

Step 1 — Bring down the old network

```
bash
cd ~/fabric-samples/test-network
./network.sh down
```

Step 2 — Set PATH (required every new terminal)

```
bash
export PATH=$PATH:$HOME/fabric-samples/bin
export FABRIC_CFG_PATH=$HOME/fabric-samples/config
```

Step 3 — Start network and create channel

```
bash
cd ~/fabric-samples/test-network
./network.sh up createChannel
```

Step 4 — Deploy chaincode

```
bash
./network.sh deployCC \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccl javascript
```

Step 5 — Set all environment variables

```
bash
export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export
CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peer
Organizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export
CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrg
anizations/org1.example.com/users/Admin@org1.example.com/msp
```



```
export CORE_PEER_ADDRESS=localhost:7051
```

Step 6 — Initialize the ledger

```
bash
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/
orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/
peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/
peer0.org2.example.com/tls/ca.crt \
  --waitForEvent \
  -c '{"function":"InitLedger","Args":[]}'
```

Step 7 — Query all assets

```
bash
peer chaincode query \
  -C mychannel \
  -n basic \
  -c '{"Args":["GetAllAssets"]}'
```

Step 8 — Read a specific asset

```
bash
peer chaincode query \
  -C mychannel \
  -n basic \
  -c '{"Args":["ReadAsset","asset5"]}'
```

Step 9 — Create a new asset

```
bash
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/
orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/
peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/
peer0.org2.example.com/tls/ca.crt \
  --waitForEvent \
  -c '{"function":"CreateAsset","Args":["asset10","blue","10","Arnav","1000"]}'
```

Verify:

```
bash
peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
```

Step 10 — Transfer asset ownership

```
bash
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/
orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/
peer0.org1.example.com/tls/ca.crt \
```

```
--peerAddresses localhost:9051 \  
--tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/  
peer0.org2.example.com/tls/ca.crt \  
--waitForEvent \  
-c '{"function":"TransferAsset","Args":["asset10","Rahul"]}'
```

Verify:

```
bash  
peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset10"]}'
```

Step 11 — Stop the network when done

```
bash  
cd ~/fabric-samples/test-network  
./network.sh down
```